

Spazi di Sobolev.

Recall: $u \in L^p(\Omega) \subseteq L^1_{loc}(\Omega) \mapsto T_u \in \mathcal{D}'(\Omega)$
 $\Omega \subseteq \mathbb{R}^n$ fissato $i \in \{1, \dots, n\}$

$$\left\langle \underbrace{\frac{\partial}{\partial x_i}(T_u)}_{\substack{\uparrow \\ \mathcal{D}'(\Omega)}}, \varphi \right\rangle = - \left\langle T_u, \frac{\partial \varphi}{\partial x_i} \right\rangle \quad \forall \varphi \in \mathcal{D}(\Omega)$$

Più brevemente:

$$\left| \left\langle \frac{\partial u}{\partial x_i}, \varphi \right\rangle \right. \quad (=) \quad \left. - \left\langle u, \frac{\partial \varphi}{\partial x_i} \right\rangle \quad \forall \varphi \in \mathcal{D}(\Omega) \right|$$

In particolare, data $u \in L^p(\Omega)$,

chiedere che $\frac{\partial u}{\partial x_i} \in L^p(\Omega)$ significa:

chiedere che $\exists v \in L^p(\Omega)$ tale che

la distribuzione $\frac{\partial}{\partial x_i}(T_u) = \frac{\partial u}{\partial x_i}$ sia

della forma T_v con $v \in L^p(\Omega)$.

ovvero

$$\boxed{\exists v \in L^p(\Omega) \text{ tale che} \\ \int_{\Omega} v \varphi = - \int_{\Omega} u \frac{\partial \varphi}{\partial x_i} \quad \forall \varphi \in \mathcal{D}(\Omega)} \quad v$$

$$\Downarrow \\ v = \frac{\partial u}{\partial x_i} \quad (\text{nel senso distribuzionale})$$

In particolare, per $n=1$: $\Omega = I$

dato $u \in L^1(I)$, $u' \in L^1(I)$ significa

$\exists v \in L^1(I)$ tale che

$$\boxed{\int_I v \varphi = - \int_I u \varphi' \quad \forall \varphi \in \mathcal{D}(I)}$$

$$\Downarrow \\ v = u' \quad (\text{nel senso distribuzionale})$$

Esemp:

- $u(x) = |x| \in L^2(I) \quad (-1,1)$
 $u'(x) = \begin{cases} -1 & x < 0 \\ 1 & x > 0 \end{cases} = \text{sign } x \in L^2(I).$
 $\uparrow$
in $\mathcal{D}'(I)$.

$$\left[(T_u)' = \int \text{sign } x \right]$$

- $u(x) = (\text{sign } x) \in L^2(I) \quad (-1,1)$
 $u'(x) \notin L^2(I)$

$$\left[(T_u)' = \frac{2}{n} \delta_0 \right] \quad L^2(\Omega)$$

Def. fissato $\Omega \text{ ap. } \subseteq \mathbb{R}^n$, e $p \in [1, +\infty]$

$$W^{1,p}(\Omega) := \left\{ u \in L^p(\Omega) : \frac{\partial u}{\partial x_i} \in L^p(\Omega) \quad \forall i=1, \dots, n \right\}$$

spazio di Sobolev

$$\boxed{n=1} : u \in W^{1,p}(\Omega) \iff u, u' \in L^p(I).$$

$$\begin{aligned} \bullet \quad u \in C_0^1(I) &\Rightarrow u, u' \in C_0^\infty(I) \subseteq L^p(I) \\ &\Rightarrow u \in W^{1,p}(\Omega). \end{aligned}$$

$$\bullet \quad u(x) = |x| \Rightarrow u, u' \in L^p(I) \Rightarrow u \in W^{1,p}(I)$$

$$\bullet \quad u(x) = \sin x \Rightarrow u \notin W^{1,p}(I)$$

Sia $k \in \mathbb{N}$, $k \geq 1$

$$W^{k,p}(\Omega) := \left\{ u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \quad \forall \alpha \text{ multi-indice con } |\alpha| \leq k \right\}$$

$$\alpha = (\alpha_1, \dots, \alpha_n) \Rightarrow |\alpha| = \alpha_1 + \dots + \alpha_n$$

$$\boxed{n=1} : u \in W^{1,p}(I) \iff u, u', \dots, u^{(k)} \in L^p(I)$$

Observation Sia $K \in \mathbb{N}$, $K \geq 1$, $p \in [1, +\infty]$

• $W^{K,p}(\Omega)$ sono spazi vettoriali.

Ad esempio $K=1$, $n=1$

$$W^{1,p}(I) = \{u \in L^p(I), u' \in L^p(I)\}$$

Siano $u, v \in W^1 p(I)$, $\alpha, \beta \in \mathbb{R}$.

Verifico $\alpha u + \beta v \in W^{1,p}(I)$.

1) $\alpha u + \beta v \in L^p(I)$

vero perché $u, v \in L^p(I)$, e so che
 $L^p(I)$ sp. vettoriale.

2) $(\alpha u + \beta v)' \in L^p(I)$

②

$$\alpha u' + \beta v'$$

vero perché $u', v' \in L^p(I)$ e so che
 $L^p(I)$ sp. vettoriale

Verifica della uguaglianza in blue

$$(\underbrace{\alpha u + \beta v}_w)' \stackrel{!}{=} \underbrace{\alpha u' + \beta v'} \text{ in } \mathcal{D}'(I)$$

per def. significa

$$\int_I (\alpha u' + \beta v') \varphi = - \int_I \underbrace{(\alpha u + \beta v)}_w \varphi' \quad \forall \varphi \in \mathcal{D}(I)$$

$\downarrow$ linearità $\int$ $\downarrow$ linearità $\int$

$$\alpha \int_I u' \varphi + \beta \int_I v' \varphi \longleftrightarrow -\alpha \int_I u \varphi' - \beta \int_I v \varphi'$$


vale per def. di u' e v' in $\mathcal{D}'(I)$

Def. Norme su $W^{1,p}(\Omega)$

$$\|u\|_{1,p} := \|u\|_p + \sum_{i=1}^n \left\| \frac{\partial u}{\partial x_i} \right\|_p$$

$$(n=1 \Rightarrow \|u\|_{1,p} := \|u\|_p + \|u'\|_p).$$

Def. $u_h \rightarrow u$ in $W^{1,p}(\Omega) \iff$

$$\|u_h - u\|_{1,p} \rightarrow 0. \iff$$

$$\begin{cases} \|u_h - u\|_p \rightarrow 0 \\ \left\| \frac{\partial u_h}{\partial x_i} - \frac{\partial u}{\partial x_i} \right\|_p \rightarrow 0 \quad \forall i=1, \dots, n. \end{cases} \iff$$

$$\begin{cases} u_h \rightarrow u \text{ in } L^p(\Omega) \\ \frac{\partial u_h}{\partial x_i} \rightarrow \frac{\partial u}{\partial x_i} \text{ in } L^p(\Omega) \quad \forall i=1, \dots, n \end{cases}$$

$$\bullet \sum_{i=1}^n \left\| \frac{\partial u}{\partial x_i} \right\|_p \rightsquigarrow \|\nabla u\|_p$$

• $n=1$ $p \in [1, +\infty)$

$$\|u\|_{1,p} = \left(\int_{\Omega} |u|^p \right)^{1/p} + \left(\int_{\Omega} |u'|^p \right)^{1/p}$$

$$\|u\|_{1,p} = \left(\int_{\Omega} |u|^p + |u'|^p \right)^{1/p}$$

sono entrambe norme su $W^{1,p}(\Omega)$

e sono norme EQUIVALENTI.

Def. Norme su $W^{k,p}(\Omega)$

$$\|u\|_{k,p} := \sum_{|\alpha| \leq k} \|D^{\alpha} u\|_p$$

$$(n=1 \Rightarrow \|u\|_{k,p} := \|u\|_p + \|u'\|_p + \dots + \|u^{(k)}\|_p)$$

o equivalentemente

$$\|u\|_{k,p} = \left(\int_{\Omega} |u|^p + |u'|^p + \dots + |u^{(k)}|^p \right)^{1/p}$$

Teorema $W^{k,p}(\Omega)$ sono Banach
muniti delle norme $\|\cdot\|_{k,p}$.

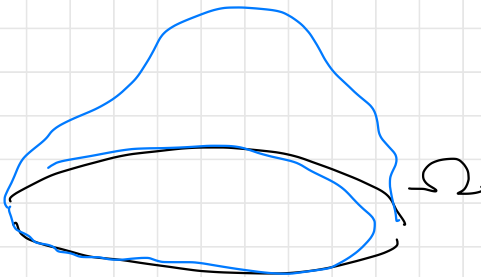
Casi particolari $(p=2)$

$$\frac{1}{p} + \frac{1}{p'} = 1$$

$$W^{1,2}(\Omega) = H^1(\Omega) = \{u \in L^2(\Omega), \nabla u \in L^2(\Omega)\}$$

$$W^{k,2}(\Omega) = H^k(\Omega) = \left\{ u \in L^2(\Omega) : \partial^\alpha u \in L^2(\Omega) \right. \\ \left. \forall \alpha \text{ con } |\alpha| \leq k \right\}$$

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{su } \partial\Omega \end{cases} \quad \text{eq. membrane}$$



$$\Delta u = u_{x_1 x_1} + u_{x_2 x_2}$$



Def.

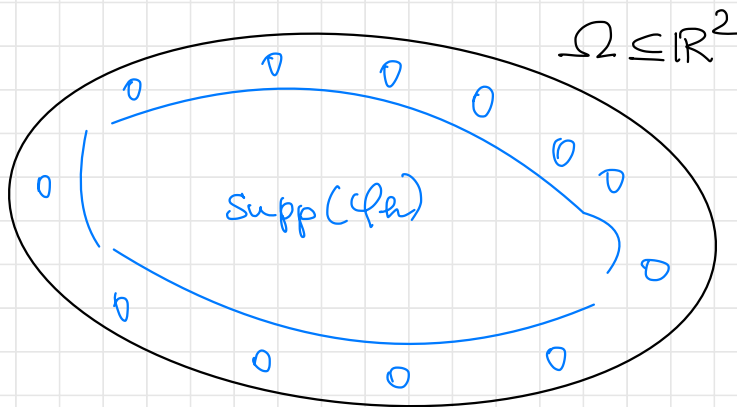
funzioni di $W^{1,p}(\Omega)$ "nulle al bordo"

$W_0^{1,p}(\Omega) :=$ chiusura di $\underbrace{C_0^\infty(\Omega)}_{\text{sottospazio di } W^{1,p}(\Omega)}$ in $W^{1,p}(\Omega) =$

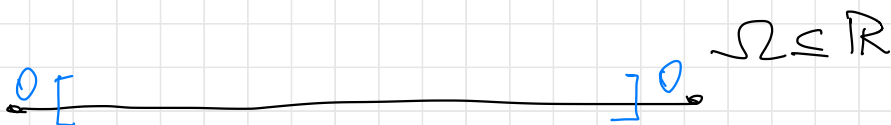
$$= \{u \in W^{1,p}(\Omega) : \exists \{\varphi_h\} \subseteq C_0^\infty(\Omega) \text{ tale che } \varphi_h \rightarrow u \text{ in } W^{1,p}(\Omega)\}$$

$$= \{u \in W^{1,p}(\Omega) : \exists \{\varphi_h\} \subseteq C_0^\infty(\Omega) \text{ tale che } \varphi_h \rightarrow u \text{ in } L^p(\Omega) \\ \forall \varphi_h \rightarrow \nabla \varphi_h \text{ in } L^p(\Omega)\}$$

- $C_0^\infty(\Omega) \subseteq W^{1,p}(\Omega)$
- $C_0^\infty(\Omega)$ sottospazio vettoriale



$\Omega \subseteq \mathbb{R}^2$



$\Omega \subseteq \mathbb{R}$

Osservazioni

- $u \in W^{1,p}(\Omega) \cap C(\bar{\Omega}) \Rightarrow$ PROPOSIZIONE
 $\underbrace{u=0 \text{ su } \partial\Omega}_{\text{in senso classico}} \text{ se e solo se } u \in W_0^{1,p}(\Omega)$
- $\triangle!$ Differente al teo di densità
di $C_0^\infty(\Omega)$ in $L^p(\Omega)$ $\triangle!$
 $\forall u \in L^p(\Omega) \exists \varphi_h \in C_0^\infty(\Omega)$ tale che $\varphi_h \rightarrow u$
 $\text{in } L^p(\Omega)$
- $W_0^{1,p}(\Omega)$ è uno spazio di Banach.
perché è un sottospazio chiuso di un Banach
 $\uparrow$ $\uparrow$
per def. $W^{1,p}(\Omega)$
- M ssp. chiuso $\subseteq V$ Banach $\Rightarrow$
 M Banach. perché:
 $\{v_m\} \subseteq M$ di Cauchy $\Rightarrow$
 $v_m \rightarrow v$ in V (perché V Banach)
ma so anche che M chiuso $\Rightarrow v \in M \quad \square$

Teorema (disuguaglianza di Poincaré in $W_0^{1,p}(\Omega)$)

Sia Ω aperto LIMITATO in $\mathbb{R}^n$.

Allora $\exists C_p(\Omega) > 0$ tale che

$$\|u\|_p \leq C_p(\Omega) \cdot \|\nabla u\|_p \quad \forall u \in W_0^{1,p}(\Omega)$$

Idea $n=1$ $u \in C^1(0,1)$ $p=2$

$$u(x) = \cancel{u(0)} + \int_0^1 u'(t) dt \Rightarrow$$

$$|u(x)| = \left| \int_0^1 u' \right| \leq \int_0^1 |u'| \leq \|u'\|_{L^2(0,1)} \quad \uparrow \text{ immersione}$$

$$\Rightarrow \forall x \in (0,1) \quad |u(x)| \leq \|u'\|_{L^2(0,1)}$$

$$\Rightarrow \forall x \in (0,1) \quad |u(x)|^2 \leq \|u'\|_{L^2(0,1)}^2$$

Integro $u(0,1)$

$$\int_0^1 |u(x)|^2 \leq \|u'\|_{L^2(0,1)}^2 \cdot \underbrace{|(0,1)|}_1$$

$$\|u\|_{L^2(0,1)} \leq \|u'\|_{L^2(0,1)}.$$

□

Conseguenza Su $W^{1,p}_0(\Omega)$,

$$\|u\|_{1,p} = \|u\|_p + \|\nabla u\|_p$$

$$\|\nabla u\|_p \quad (\text{è una norma!})$$

sono equivalenti!

$$\|\nabla u\|_p \leq \|u\|_{1,p}$$

$$\|u\|_{1,p} = \underbrace{\|u\|_p}_{\leq C_p(\Omega) \|\nabla u\|_p} + \|\nabla u\|_p$$

$$\leq \underbrace{C_p(\Omega)}_{\text{costante}} \|\nabla u\|_p + \|\nabla u\|_p$$

$$= \underbrace{(C_p(\Omega) + 1)}_{\text{costante}} \|\nabla u\|_p$$